COMPUTER NETWORKS LAB REPORT

Experiment Title

Bandwidth Measurement and Analysis using Mininet and iperf

Student Details

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1. AIM

To measure and analyze network bandwidth under different configurations using Mininet and iperf.

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2. OBJECTIVES

- To install and configure Mininet on Ubuntu
- To create virtual network topologies
- To measure throughput using iperf
- To compare performance under different network conditions

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3. SYSTEM REQUIREMENTS

- Ubuntu OS (20.04/22.04)
- Mininet
- iperf tool
- Terminal access

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4. THEORY

Mininet is a network emulator that creates a virtual network on a single machine using lightweight virtualization. It allows testing of network protocols and configurations.

iperf is a command-line tool used to measure network performance, especially bandwidth and throughput between two hosts.

5. INSTALLATION AND SETUP

Mininet was installed using the following commands:

sudo apt update
sudo apt install mininet -y

```
manjuntha@manjuntha-VirtualBox:~$ sudo apt update
sudo apt install mininet -y
[sudo] password for manjuntha:
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,402 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadata [54.5 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 DEP-11 Metadata [208 B]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 DEP-11 Metadata [125 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 DEP-11 Metadata [208 B]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [114 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 DEP-11 Metadata [212 B]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe i386 Packages [801 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,262 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 DEP-11 Metadata [360 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy-backports/main amd64 DEP-11 Metadata [7,352 B]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 DEP-11 Metadata [212 B]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 DEP-11 Metadata [12.4 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 DEP-11 Metadata [212 B]
Fetched 6,524 kB in 5s (1,371 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
184 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mininet is already the newest version (2.3.0-1ubuntu1).
The following package was automatically installed and is no longer required:
  libreoffice-ogltrans
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 184 not upgraded.
manjuntha@manjuntha-VirtualBox:~$
```

iperf was installed using:

sudo apt install iperf -y

```
manjuntha@manjuntha-VirtualBox:~$ sudo apt install iperf -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
iperf is already the newest version (2.1.5+dfsg1-1).
iperf set to manually installed.
The following package was automatically installed and is no longer required:
  libreoffice-ogltrans
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 184 not upgraded.
manjuntha@manjuntha-VirtualBox:~$
```

6. PROCEDURE

Step 1: Start Mininet

sudo mn

```
manjuntha@manjuntha-VirtualBox:~$ sudo mn
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller

*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

Step 2: Test Network Connectivity

pingall

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet>
```

Step 3: Create Custom Topology

sudo mn --topo single,3

```
manjuntha@manjuntha-VirtualBox:~$ sudo mn --topo single,3
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller

*** Starting 1 switches
```

Step 4: Start iperf Server

h1 iperf -s &

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7. EXPERIMENTAL RESULTS

Scenario 1: Normal Network

h2 iperf -c h1

```
mininet> h1 iperf -s &
mininet> h2 iperf -c h1
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 37014 connected with 10.0.0.1 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0068 sec  43.6 GBytes  37.4 Gbits/sec
mininet>
```

Observed Bandwidth: 37.4 Gb/s

Scenario 2: Parallel Streams

h2 iperf -c h1 -P 4

```
mininet> h2 iperf -c h1 -P 4
[  1] local 10.0.0.2 port 38580 connected with 10.0.0.1 port 5001
[  3] local 10.0.0.2 port 38598 connected with 10.0.0.1 port 5001
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  4] local 10.0.0.2 port 38600 connected with 10.0.0.1 port 5001
[  2] local 10.0.0.2 port 38590 connected with 10.0.0.1 port 5001
[ ID] Interval       Transfer     Bandwidth
[  4] 0.0000-10.0030 sec  13.2 GBytes  11.3 Gbits/sec
[  1] 0.0000-10.0029 sec  10.5 GBytes  9.04 Gbits/sec
[  3] 0.0000-10.0041 sec  9.82 GBytes  8.44 Gbits/sec
[  2] 0.0000-10.0147 sec  12.4 GBytes  10.7 Gbits/sec
[SUM] 0.0000-10.0069 sec  46.0 GBytes  39.5 Gbits/sec
[ CT] final connect times (min/avg/max/stdev) = 5.907/7.722/9.715/2.097 ms (tot/err) = 4/0
mininet>
```

Observed Bandwidth: 39.5 Gb/s

Scenario 3: Network Delay

sudo mn --topo single,3 --link tc,delay=10ms

Then:

h1 iperf -s &

h2 iperf -c h1

```
manjuntha@manjuntha-VirtualBox:~$ sudo mn --topo single,3 --link tc,delay=10ms
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
```

Observed Bandwidth: 1.32 Gb/s

Scenario 4: Limited Bandwidth

sudo mn --topo single,3 --link tc,bw=10
Then:

h1 iperf -s &
h2 iperf -c h1

```
manjuntha@manjuntha-VirtualBox:~$ sudo mn --topo single,3 --link tc,bw=10
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(10.00Mbit) (10.00Mbit) (h1, s1) (10.00Mbit) (10.00Mbit) (h2, s1) (10.00Mbit) (10.00Mbit) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller

*** Starting 1 switches
s1 ...(10.00Mbit) (10.00Mbit) (10.00Mbit)
*** Starting CLI:
mininet> h1 iperf -s &
mininet> h2 iperf -c h1
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size:  128 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 60838 connected with 10.0.0.1 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-11.5923 sec  13.0 MBytes  9.41 Mbits/sec
mininet>
```

Observed Bandwidth: 9.41 Mb/s

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8. OBSERVATION TABLE

Scenario	Bandwidth	Observation
Normal	37.4 Gb/s	Stable performance
Parallel	39.5 Gb/s	Increased throughput
Delay	1.32 Gb/s	Reduced due to latency
Limited	9.41 Mb/s	Bandwidth restricted

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9. ANALYSIS

- Bandwidth varies depending on network configuration
- Parallel streams improve bandwidth utilization
- Network delay reduces effective throughput
- Bandwidth limiting enforces strict speed constraints
- Mininet effectively simulates real-world network behavior

10. RESULT

The experiment successfully measured and compared bandwidth under various network conditions using Mininet and iperf.

11. CONCLUSION

Mininet and iperf provide an efficient and flexible environment for testing and analyzing network performance. The experiment demonstrated how different parameters affect bandwidth.

12. REFERENCES

- Mininet Official Documentation
- iperf Manual